



Which social factors best predict maternal morbidity across U.S. states?

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01

Project Motivations

Maternal Morbidity

➤ ICU Admission

- Severe Postpartum Hemorrhage
- Hypertensive Disorders
- Sepsis

➤ Transfusion

- Heavy bleeding (hemorrhage) during delivery
- Ruptured ectopic pregnancy

➤ Perineal Laceration

- Tear in the skin and other tissues that happen during vaginal birth

➤ Unplanned Hysterectomy

- Emergency removal of the uterus during another surgery
- uncontrollable bleeding or organ damage

➤ Uterine Rupture

- Rare and dangerous tear in the muscular wall of the uterus

Definition

- Short- or long-term physical or mental health problem, illness, or disability that results from or is aggravated by pregnancy, childbirth, or the postpartum period

Which social factors best predict maternal morbidity across U.S. states?



Address

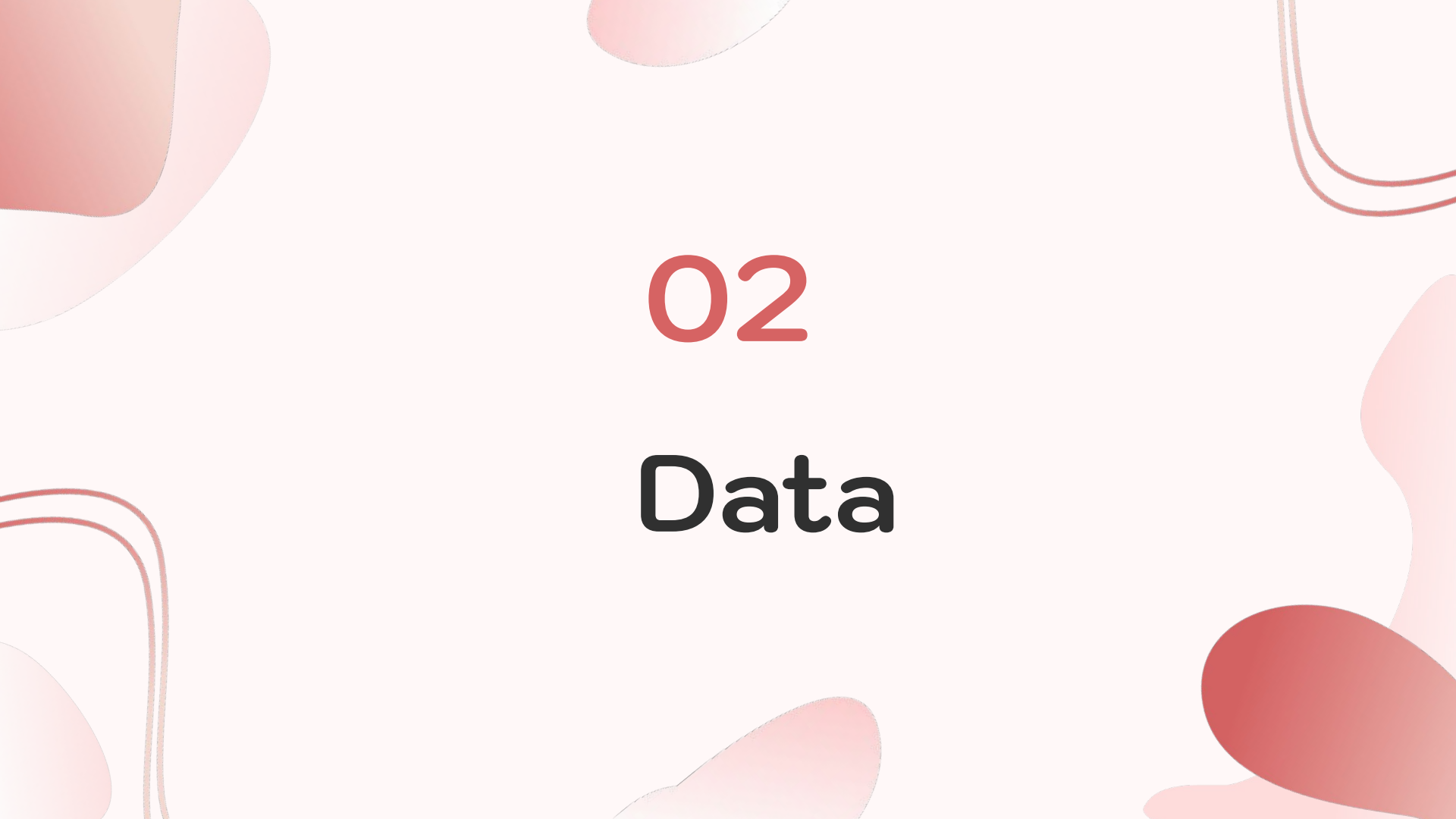
Social Determinants of Health:

- Race
- Ethnicity
- Education
- Age
- Insurance (income)
- Urban/rural living (access to OBGYN)



Importance

Determining which of these factors carries the greatest independent weight is not just an academic question, instead, it can directly inform where policymakers and public health officials should prioritize limited resources to reduce preventable maternal morbidity and mortality



02

Data

CDC Wonder

Query 1: Births by State x Race x Age x Education x Maternal Morbidity Checked

Query 2: Births by State x Urbanicity (Metro/Nonmetro) x Hispanic Origin x Source of Payment for Delivery x Maternal Morbidity Checked



03

Initial Findings

Initial Methods (n = 51, state-level)



OLS Coefficient ($R^2 = .373$)

- Each coefficient represents the estimated change in a state's morbidity rate associated with a one-standard-deviation increase in that predictor, holding all other predictors constant.
- Only one variable, % **teen births**, was statistically significant (coefficient = -0.0053, $p = 0.03$)
 - meaning states with more teen births tended to have *lower* overall morbidity rates, holding everything else constant.

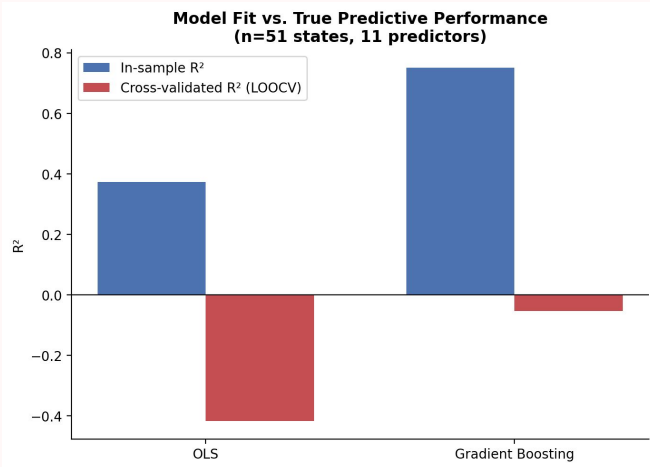


Gradient Boosting ($R^2 = .752$)

- more flexible model than OLS
- **Overfitting:** Memorized the specific states rather than establishing a pattern

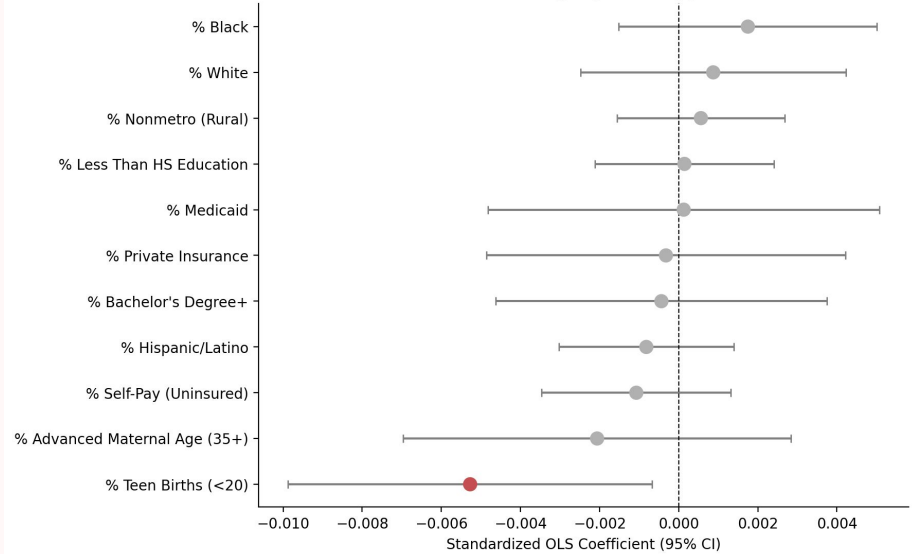


Findings



- Gradient Boosting has a much higher R^2
- When both are cross-validated, they are both negative → they both failed to create a model that accurately predicted morbidity rates

Effect of Each Factor on State-Level Maternal Morbidity Rate (red = statistically significant, $p < 0.05$)



- OLS Coefficient: the only variable that was statistically significant was **% teen births**.
- Every other coefficient had p-values above 0.05
- 51 states and 11 predictors, the model didn't have enough data relative to its complexity



04

Final Findings

Final Analytical Methods

CatBoost



- ❖ Query 1 (AUC = 0.82) confirming logistic regression result rather than dramatically outperforming it
 - **SHAP importance chart:** ranks entire *variables* by how much they influence predictions overall.
 - Race was the strongest predictor
- ❖ Query 2 (AUC = 0.49-0.52) showed much weaker performance in both logistic regression and CatBoost → no individual-level predictive signal from these variables.



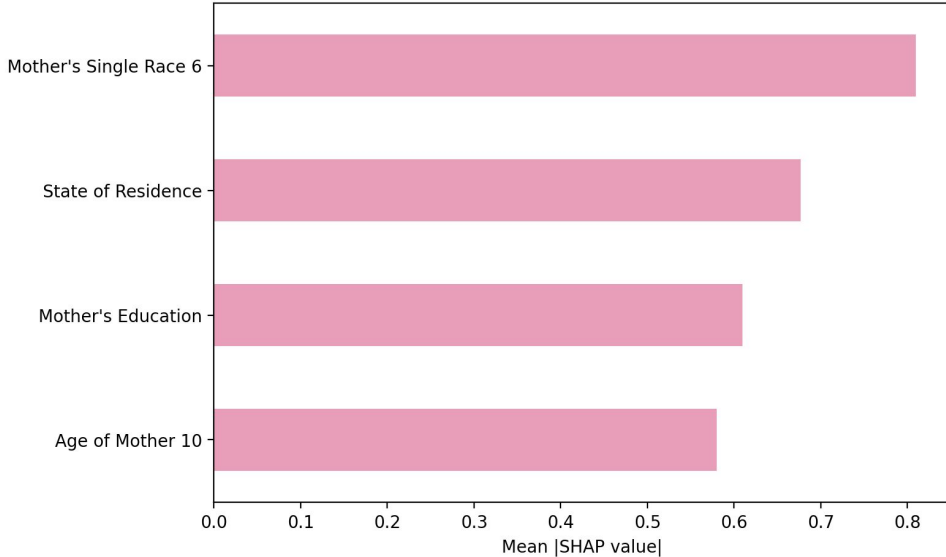
OLS Coefficient: do these factors work the same way in every state?

- ❖ 5 states: Texas, California, New Hampshire, Alabama, Nebraska

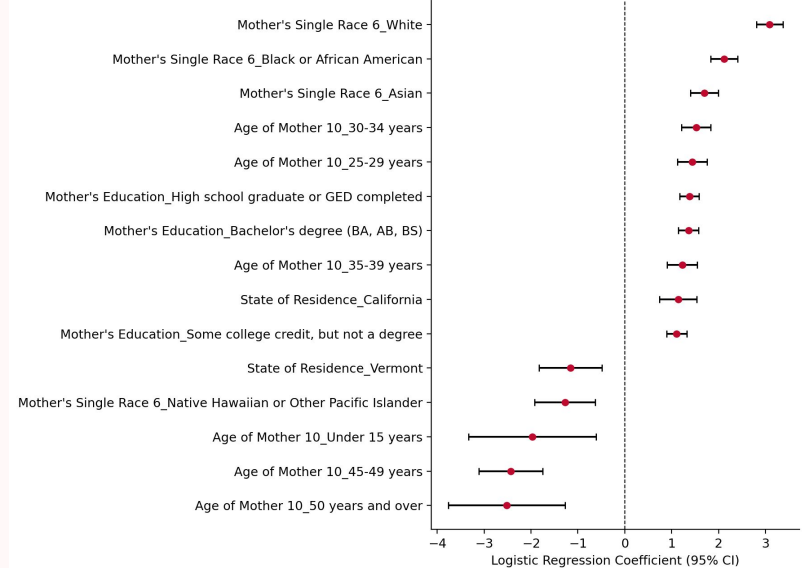
Q1 Findings (Race x Age x Education)

- Columns: 14, 561

CatBoost Feature Importance (SHAP)



Query 1 Forest Plot: Top 15 Predictors of Maternal Morbidity

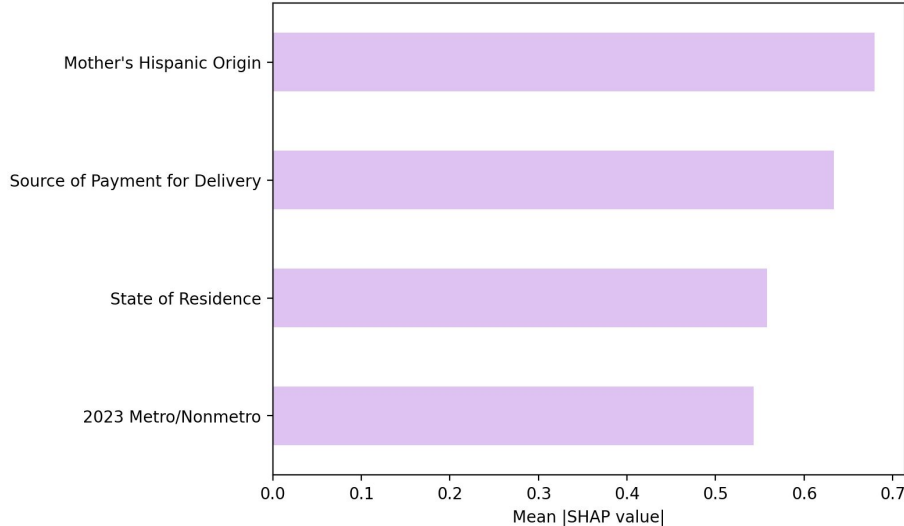


- On an individual level, race is the strongest predictor for maternal morbidity, for Q1.
- Births in the White category had the highest predicted probability of a *documented* morbidity, relative to the reference group.

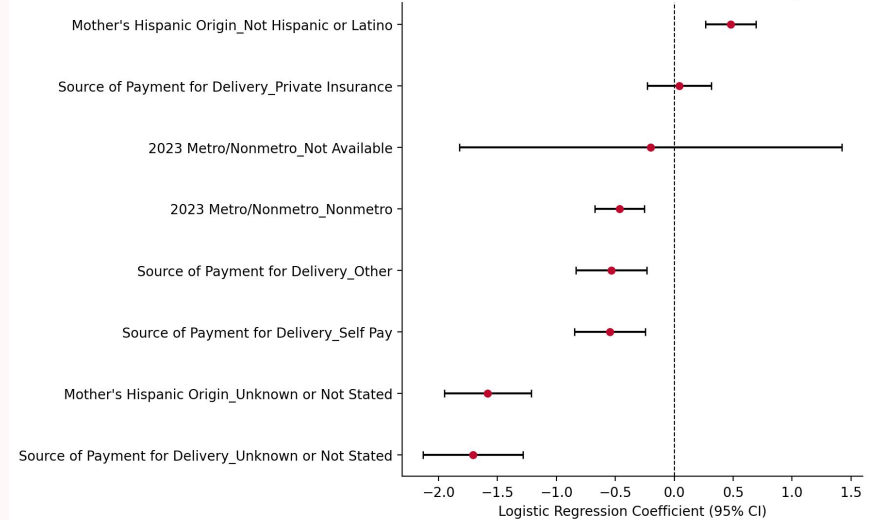
Q2 Findings (Ethnicity x Insurance x Urbanacity)

Columns = 1,939

Query 2 CatBoost Feature Importance (SHAP)

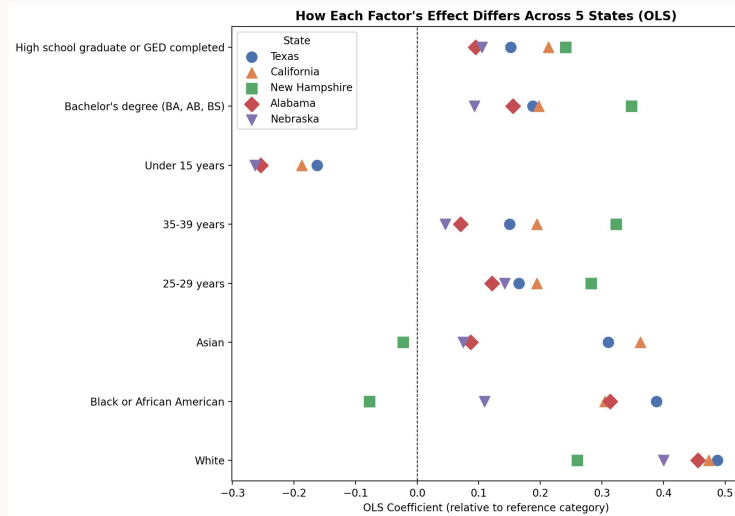


Query 2 Forest Plot: Urbanicity, Hispanic Origin, Insurance
(State dummies excluded for readability)



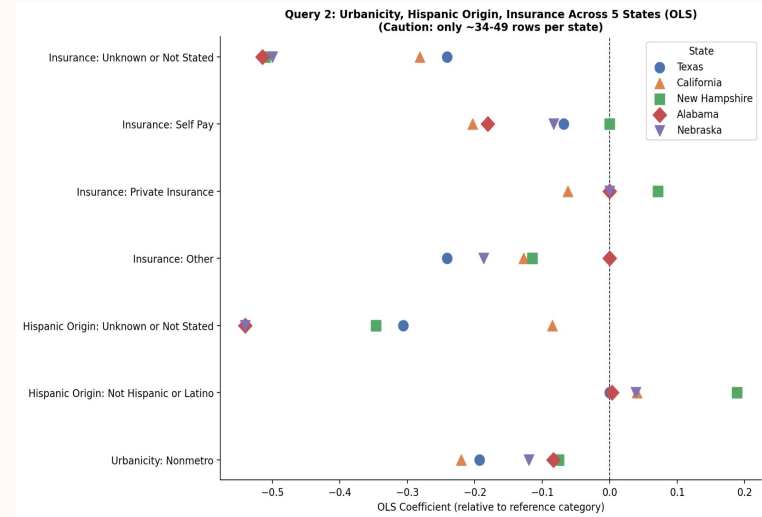
- Q2 has a much weaker predictive performance
- On an individual level, Hispanic origin, is the strongest predictor for maternal morbidity, for Q2.
- Query 2 simply has far fewer distinct combinations for a model to learn from compared to Q1, despite having roughly the same amount of births to account from (~12.5 million)

State-level Data



Query 1

some effects are fairly consistent in direction across states, while others vary considerably, suggesting state-specific context matters.



Query 2

some effects are fairly consistent in direction across states, while others vary considerably, suggesting state-specific context matters.

Severe Maternal Morbidity Across 5 Focus States





05

Conclusion

Conclusion

Challenge

OLS and gradient boosting failed to generalize at the state level (negative cross-validated R^2)

- revealing that 51 observations cannot reliably support many predictors

Lessons

"Morbidity checked" is a documentation-dependent measure, not a direct clinical outcome.

Patterns that look like demographic risk differences may partly reflect differences in how thoroughly complications get recorded across hospitals and states, not solely true differences in underlying risk

Findings

The "most important factor" changes depending on the level of analysis:

- state-aggregate level = **teen birth rate** most dominant factor
- individual/demographic-combination level = **race** most prominent factor

Displays how ecological (state-level) and individual-level patterns can diverge

Summary

No single social determinant dominates maternal morbidity outcomes uniformly, as the answer shifts with scale (state vs. individual), underscoring that policy interventions likely need to be tailored rather than built around one universal "top factor."